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A novel model-based welding trajectory planning method for identical structural workpieces

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ABSTRACT

Welding robots have been widely used with the development of manufacturing industry. At present, welding trajectory planning and programming of welding robots are performed separately for similar (structural) workpieces with different dimensions or deformations. Besides, the welding torch pose planning using conventional robot programming methods is time-consuming for workpieces with complex welding trajectories. To realize automatic planning of the welding torch pose for similar workpieces and improve the programming efficiency of welding robots, we propose a novel model-based welding trajectory planning method for welding robots. We create a basic model (point set) containing information on welding seam trajectory planning for similar workpieces. The welding seam trajectory extraction of similar workpieces is achieved by the non-rigid point set registration of the feature point cloud of similar workpieces with the basic model. Then, the welding torch pose planning is implemented by the geometric projection method of the local point cloud of the welding seam trajectory. Finally, the results of offline experiments and online experiments demonstrate the effectiveness of the proposed method.

1. Introduction

With the continuous development of industry, industrial robots are widely used in various scenarios, such as palletizing, welding, assembly, inspection, and surface processing. Industrial robots have advantages such as higher accuracy, better reliability, productivity, and safety compared to humans. In practical industrial applications, there are two main types of programming methods for welding robot applications: online programming methods and offline programming methods [1]. In online programming, the operator usually controls the robot through a teach pendant to bring its end-effector to the designated position, records the robot's pose data and writes instructions for the robot's motion, and then generates the robot's program [2]. Compared to online programming, offline programming does not require robot downtime and can be implemented in a computer-built virtual environment using the workpiece's 3D model. Welding tasks with complex welding trajectories require the realization of pose planning for the welding torch, which is relatively time-consuming and demanding for the operator by using either online or offline programming methods.

In recent years, many scholars have studied the planning of robot end-effector to improve the programming efficiency of robots. According to the way when the robot acquires the workpiece information or

task information, the robot end-effector pose planning can be categorized into online method, offline method, and hybrid method. In online method, scholars utilize different sensors to perceive the working environment, and process the perceived information through corresponding algorithms to realize the robot end-effector pose planning. Matteo [3] proposed an approach to lead-through robot programming based on the manipulator dynamic model and proprioceptive measures. Instead of using the robot teach pendant, a more complex dynamic model was needed to make it easier for the operator to control the robot. Hügler [4] proposed a novel approach towards hybrid programming of industrial robots through a combination of haptic and non-haptic gestural interaction. Further, to make non-expert users intuitively and efficiently program trajectories for robots, Zhang proposed a new trajectory programming system that tracks a pen-shaped marker and obtains its position and orientation by processing the point cloud data of the workspace [5]. Zhang proposed a solution for programming industrial robots based on programming by demonstration (PbD) [6]. The robot's motion information is obtained from the demonstration process of a human performing the task. Fabian [7] recorded the welders' simulated welding process through vision sensors and converted the welding trajectory obtained from the recording into an executable robot code to complete the programming of the robot. Besides, Geng proposed a

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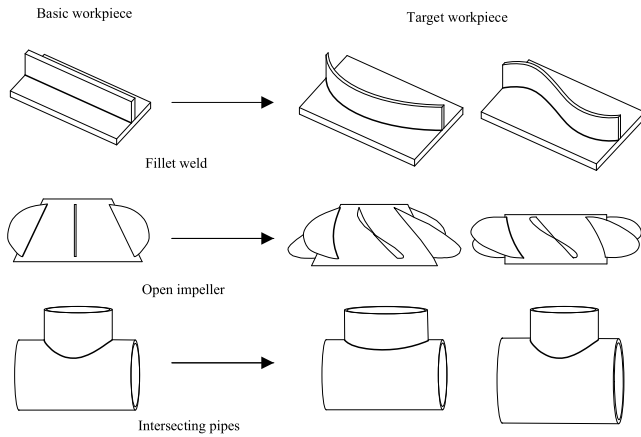


Fig. 1. Schematic diagram of typical welding workpieces.

3D vision-based seam extraction and path planning method for robotic welding of medium-thick plate structural parts [8]. Compared with online methods, the 3D model of the workpiece is generally available when using offline methods to realize trajectory planning. Therefore, researchers were more concerned about how to utilize the 3D model of the workpiece to improve the efficiency of robot programming and reduce the difficulty of robot end-effector pose planning. Chen proposed to obtain workpiece and their manufacturing programs similar to the target workpiece by histogram method and generate the surface machining program method of the target workpiece by certain adjustments based on the existing program database and the corresponding 3D model of the workpiece [9]. Neto proposed a novel system for CAD-based off-line robot programming, robot programs are directly generated from a 3-D CAD drawing running on a commonly available 3-D CAD package [10,11]. Bedaka designed and develop an offline programming simulation platform for welding applications based on OPEN CASCADE libraries in C++ integration environment [12]. Tran proposed a CAD integrated method for automatic recognition of potential weld locations in large assembly structures predominantly comprised of weld joints [13]. To accommodate more complex robot end-effector planning, scholars have fused online and offline methods to achieve more efficient and intelligent robot programming. A novel approach using augmented reality (AR) to facilitate robot programming in an unknown environment where the user directly interacts with a virtual robot was proposed [14]. Ong presented an augmented reality-assisted robot welding task programming (ARWP) system using a user-friendly AR interface that simplifies and speeds up the programming of robotic welding tasks [15]. This type of method can reduce the professional quality of the user and can reduce the difficulty of programming the robot. Tudor [16] proposed an approach to generic robot programming by leverages the newest graphical user interface automation tools and techniques to generate robot programs. But the utility of the method in an industrial context has not been evaluated. Zhang [17] proposed a hybrid offline programming method, systematically combining CAD-based, vision-based, and vision & CAD interactive activities, to overcome the limitations of current automatic program generation methods for robotic welding systems.

In industry, companies or factories often have multiple sizes of the same type of product for different applications or different needs. As a result, the similar (structurally identical) welding workpiece may have different dimensions or have different degrees of deformation. As shown in Fig. 1, the diagram of several typical welding workpieces. Currently, it is often necessary to plan the welding torch separately for welding workpieces with similar structures but different sizes or deformations. In the case of simple welding workpieces, the welding robot can realize the welding task by offline programming or online programming methods. However, in the case of workpieces with complex

welding seam trajectories, it is necessary not only to program complex spatial paths but also to plan the welding torch pose. In this case, the existing offline programming and online programming methods for welding robots are still time-consuming. When the similar workpieces have a small degree of deformation and the deviation of the welding seam trajectory is small, the welding seam tracking method can be used to realize the adjustment of the welding seam trajectory. However, the seam tracking method can only obtain the position information of the welding seam, so when the deformation of the same kind of workpiece is relatively large, it is difficult to adjust the welding torch pose, which affects the welding quality. In addition, the welding trajectory and planning information used to realize the welding torch pose planning requires human involvement. Hence, automatic realization of welding torch pose planning is still a challenge for welding robot programming.

To make the welding seam trajectory planning easier and improve the programming efficiency of the welding robot for similar workpieces, we propose a novel model-based welding trajectory planning method for welding robots. First, we propose a general method to generate a basic model (point set) containing workpiece structure information, welding seam trajectories and welding seam planning auxiliary trajectories. Based on the basic model, the proposed method achieves the welding seam trajectory extraction of similar workpieces by non-rigid point set registration. Then, the welding torch pose planning is achieved by the geometric relationship between the local point cloud and the welding seam trajectory. Finally, we carry out offline programming experiments and online programming experiments of the welding robot to verify the effectiveness of the proposed method.

This paper is organized as follows: The system architecture is introduced in Section 2. The proposed method is described in Section 3. The corresponding experimental results and discussion are given in Section 4. The conclusions are then given in Section 5.

2. System architecture

A typical welding robot system is shown in Fig. 2. It consists of a welding robot, a robot controller, and a welding machine. Since the proposed method is based on the structural similarity of similar workpieces for welding seam trajectory extraction, we choose to represent the structural information of the workpiece in the form of a point cloud. The point cloud of the target workpiece can be acquired using a vision sensor in the absence of the CAD file. In general, we can simplify the welding seam trajectory to produce a workpiece with the same structure as the workpiece with a complex welding seam trajectory. Therefore, we refer to a workpiece with a relatively simple welding seam trajectory as a basic workpiece of its type. The 3D model of the basic workpiece is also relatively easy to generate. However, workpieces with complex welding seam trajectories usually have a curved structure.

A model-based welding trajectory planning method is proposed for similar workpieces and to avoid repetitive programming of similar welding workpieces. Fig. 3 shows the flow chart of the proposed method. The whole process can be divided into four parts: the first step is to build the basic model, the second step is the model registration, the third step is the welding seam trajectory fitting, and the fourth step is the welding torch pose planning.

In general, we choose a simple basic workpiece as a model for a certain type of workpiece. To establish the basic model (point set), the welding seam is extracted from the 3D model of the basic workpiece using an offline programming approach. We extract the welding seam auxiliary trajectory based on the welding seam trajectory, and the auxiliary trajectory can be used to realize the pose planning of the welding torch. After the extraction of the welding seam trajectory and auxiliary trajectory of the basic workpiece is completed, the 3D model of the basic workpiece is point clouded to generate the feature point set of the basic workpiece. The point set generated by the combination of the feature point set of the basic workpiece, the welding seam

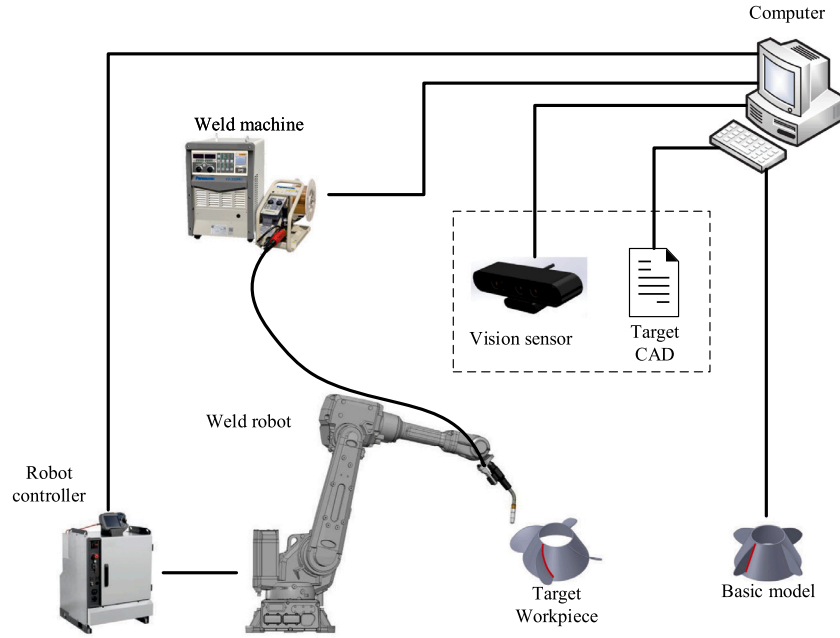


Fig. 2. The welding robot system.

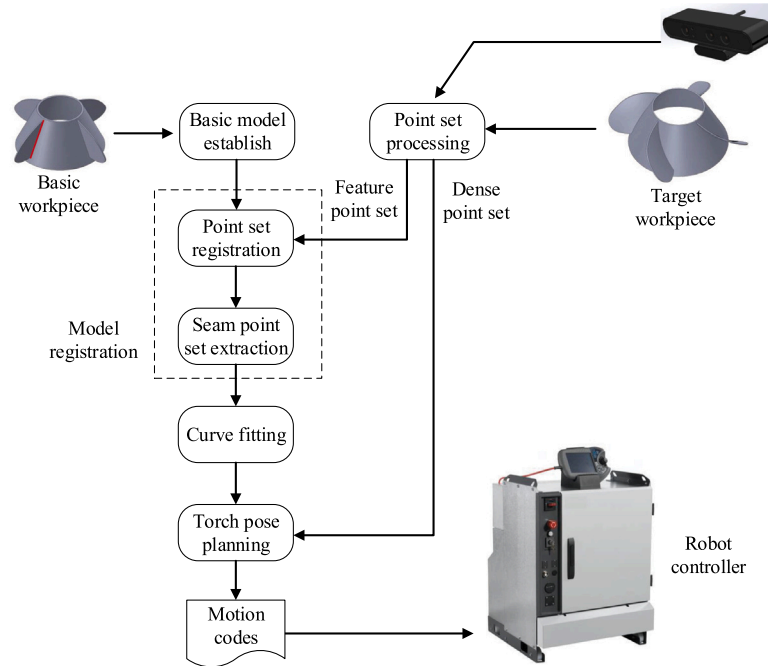


Fig. 3. The flow chart of the proposed method.

trajectory, and the auxiliary trajectory constitute the basic model of this type of workpiece. The basic model is aligned with the feature point set of the target workpiece to obtain the welding seam trajectory point set of the target workpiece. After the registration is completed, the auxiliary trajectory point set is transformed according to the same non-rigid transformation as the welding seam trajectory point set to generate the auxiliary welding seam trajectory point set of the target workpiece. To make the welding seam trajectory smoother so that it can be used for robot programming, curve fitting is performed on the obtained welding seam trajectory point set of the target workpiece and the auxiliary trajectory point set. Finally, the pose planning of the robot welding torch is implemented by using the welding seam trajectory

point set, the auxiliary trajectory point set, and the dense point cloud of the target workpiece.

3. Methodology

3.1. Basic model establish

Generally, when using a welding robot to perform a welding task, it is necessary to obtain information about the welding path and then plan the welding torch orientation according to the structure of the welding workpiece. Therefore, the welding path and structural information of the workpiece to be welded must be known for the programming tasks of the welding robot. Although the welding seam trajectory can be

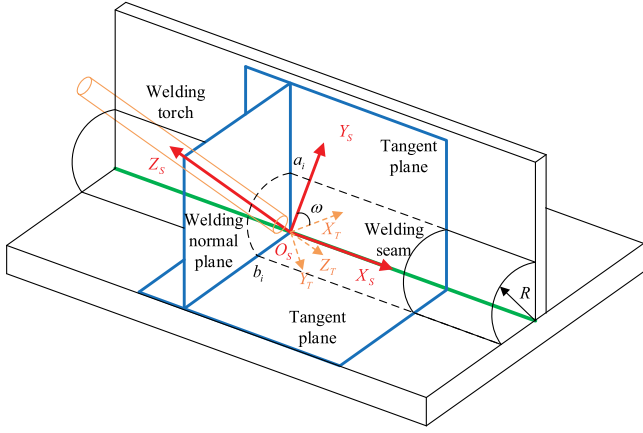


Fig. 4. Typical welding torch pose planning for fillet welding workpiece.

obtained from the 3D model of the workpiece, the welding torch pose planning is tedious as the workpiece structure becomes more complex. For similarly structured workpieces, we can generally obtain simpler basic workpieces by simplifying the welding seam trajectory, such as converting surfaces in a workpiece to planes. We select the fillet welding workpiece as an example to describe the establishment of the basic model, where the basic workpiece schematic is shown in Fig. 1.

To better describe the establishment of the basic model, we first briefly introduce the knowledge of welding torch pose planning for the basic workpiece. A typical welding torch pose planning diagram is shown in Fig. 4, where the workpiece coordinate system is defined as $X_W Y_W Z_W$; Define the current welding coordinate system $X_S Y_S Z_S$ with the current welding seam point p_i as the origin O_S , the X_S direction is the same as the welding seam trajectory forward direction, and the Z_S axis direction is the same as the angle bisector direction of the tangent plane on both sides of the current welding seam point. We define the plane with X_S axis as the normal vector and passing through the point O_S as the welding normal plane. The welding torch coordinate system is defined as $X_T Y_T Z_T$, the direction of the Z_T axis is opposite to the direction of the Z_S axis, and the angle between X_T and Y_S axis is defined as ω .

In general, the Z_S axis needs to be determined by the normal vectors of the tangent planes on both sides of the welds, but in the case of complex surfaces, the normal vectors of the tangent planes on both sides are difficult to obtain. Therefore, we adopt a general approximation to calculate the direction of the Z_S axis. The welding seam trajectory extraction can be obtained from the 3D model of the basic workpiece. Then, we can create a circular tube surface of radius R with the welding seam trajectory as the central axis, and the intersection line between this surface and the workpiece solid can be used to construct an approximate tangent plane to the current welding seam point. The surface radius of the circular tube is empirically set to 5 mm. As shown in Fig. 4, the intersection of the intersection lines with the current welding normal plane are defined as a_i and b_i , respectively. Hence, the axes of the current welding coordinate system can be solved by the following equation

$$\begin{cases} X_S = \frac{\vec{p_i p_{i+1}}}{|\vec{p_i p_{i+1}}|} \\ Z_S = \frac{\vec{p_i a_i} + \vec{p_i b_i}}{|\vec{p_i a_i} + \vec{p_i b_i}|} \\ Y_S = Z_S \times X_S \end{cases} \quad (1)$$

where p_{i+1} represents the next welding seam point.

It is easy to find from Eq. (1) that in the case of known welding seam trajectory, only Z_S axis information needs to be obtained to complete the establishment of the current welding seam coordinate system. Therefore, we can define an auxiliary point p'_i for the current

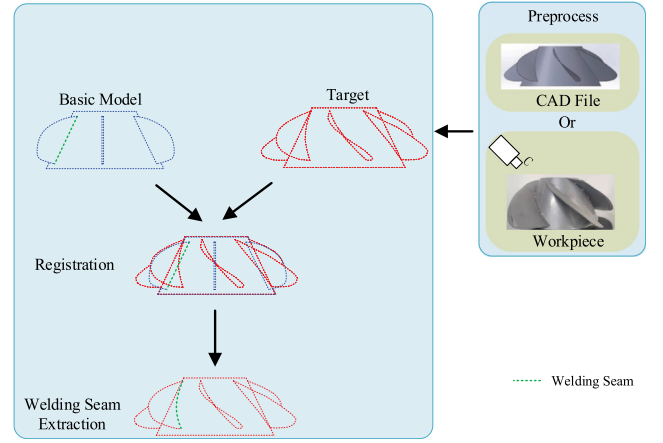


Fig. 5. Model registration schematic diagram.

welding seam point p_i . Point p'_i is located on the Z_S axis and the distance between p'_i and p_i is defined as l_p . Here, we set l_p to 15 mm based on experience. The point set $P = \{p_1, p_2, \dots, p_{M_1}\} \in \mathbb{R}^3$ is generated after the point clouding of the welding seam trajectory, and the corresponding auxiliary trajectory point set $P' = \{p'_1, p'_2, \dots, p'_{M_1}\} \in \mathbb{R}^3$ can be obtained accordingly. The point set generated by the edges of the basic workpiece excluding the welding seam trajectory is denoted as W . The basic model consists of the three point sets P , P' , and W .

3.2. Model registration

Due to the different requirements, similar workpieces remain the same in structure, but the workpieces have different dimensions or have some deformation. The idea of model registration is to extract the welding seam trajectory of the target workpiece by aligning the basic model with the target model based on the structural similarity of workpieces. With the method in Section 3.1, we can acquire the basic model. The feature point cloud of the target workpiece can be obtained by 3D model transformation or captured directly by the vision sensor. Therefore, the registration based on the structural similarity of the model can be transformed into the registration of the point cloud. Fig. 5 shows the schematic diagram of the model registration process.

In the research of this paper, the point cloud of the target workpiece is similar to the point set of the model workpiece in the main structure, but there is a certain deformation. More precisely, model registration translates into a typical non-rigid point set registration problem. The problem of point set registration (PSR) is one of the key research directions of computer image processing and medical image processing. At present, non-rigid point set registration methods are mainly divided into probability-based methods and deep learning methods [18,19]. Compared with deep learning methods, probability-based methods do not require a large number of samples for data training, and the generalization ability of the algorithm is relatively strong. Hence, we choose a probability-based non-rigid point set registration method. A typical registration algorithm based on probabilistic methods is the coherent point drift algorithm (CPD) [19]. In this paper, a Bayesian coherent point drift method is applied to achieve the registration of the target workpiece point set and the basic model point set.

We define the feature point set of the target workpiece as $X = \{x_1, x_2, \dots, x_N\} \in \mathbb{R}^3$. The model point set can be defined as $Y = \{y_1, y_2, \dots, y_M\} \in \mathbb{R}^3$ consisting of the point sets P and W . Hence, the point set registration problem can be formulated as finding a similar transformation f_{sim} and a deformation matrix V corresponding to the non-rigid deformation. A mathematical description of the registration problem can be written as

$$(\hat{V}, \hat{f}_{sim}) = \arg \min_{(V, f_{sim})} Y(V, f_{sim}, X, Y) \quad (2)$$

where Y is a loss function evaluating the dissimilarity between X and $T = f_{sim}(Y + V)$. Further, the above problem can be formulated as finding a mapping that can transform and deform the model point set to generate the target point set, for a point x_n in the target point set, the following mapping is present

$$x_n = \Gamma(y_m) = T(y_m + v_m) = sR(y_m + v_m) + t \quad (3)$$

where s is the scale factor, R is the rotation matrix, t is the translation vector, and v_m is the non-rigidly aligned displacement vector.

Based on the theory of probabilistic methods, the target point set and the model point set are considered as two probability models. The Gaussian mixture model (GMM) is introduced to invisibly encode the unknown correspondence between the points in the two point sets. The maximum likelihood function is used to solve for the unknown non-rigid transformation parameters f_{sim} and V . The probability density function can be written as:

$$p(x_n, e_n | y, v, \alpha, \rho, \sigma^2) = \prod_{m=1}^M (\alpha_m \phi_{mn})^{\delta_m(e_n)} \quad (4)$$

where $\theta = (v, \alpha, e, \rho, \sigma^2)$ is the potential variable that regulates the generation of X given Y . v is the displacement vector that characterize a non-rigid transformation of y . To express the correspondence between x_n and y_m is introduced the indicator variables e_n , where $e_n = m$ denotes the n point in the target point set corresponds to the m point in the model point set; α_m denotes the probability of correspondence with x_n and y_m ; ϕ_{mn} is an abbreviation of GMM $\phi(x_n; \Gamma(y_m); \sigma^2 I_D)$, where I_D is a identity matrix of order D ; $\rho = (s, R, t)$ denotes the similar transformation matrix of any variable.

Motion coherence is defined as the prior distribution of the displacement vector, and the prior distribution of v is defined as

$$p(v|y) = \phi(v; 0, \lambda^{-1} G \otimes I_D) \quad (5)$$

where λ is a trade-off parameter, G is a positive definite kernel matrix with values of $g_{mm'} = \exp\left(-\frac{\|y_m - y_{m'}\|^2}{2\beta^2}\right)$, β controls the directional correlation between displacement vectors. Besides, the $p(\alpha)$ defined as a Dirichlet distribution, which can be written as

$$p(\alpha) = \text{Dir}(\alpha | \kappa \mathbf{1}_M) \quad (6)$$

where $\kappa > 0$ is the parameter that controls the shape of the Dirichlet distribution. $\mathbf{1}_M$ is a M column vector of all ones. Hence, the full joint distribution $p(x, y, \theta)$ can be written as

$$p(x, y, \theta) \propto p(v|y)p(\alpha) \prod_{m=1}^M (\alpha_m \phi_{mn})^{\delta_m(e_n)} \quad (7)$$

According to the Bayesian formula, we can compute the posterior distribution of θ according to the following equation

$$p(\theta | (x, y)) = \frac{p(x, y, \theta)}{p(x, y)} \quad (8)$$

where $p(x, y)$ cannot be calculated directly. Hence, the variational Bayesian inference (VBI) method is applied to estimate the parameters θ . Typically, we can express the VBI by the minimization of the Kullback-Leibler (KL) divergence between $q(\theta)$ and $p(\theta | (x, y))$, which can be written as:

$$\hat{q}(\theta) = \arg \min_q \left\{ - \int q(\theta) \ln \left(\frac{p(\theta | (x, y))}{q(\theta)} \right) d\theta \right\} \quad (9)$$

The equivalent definition suitable for deriving the general solution is the maximization of the lower bound $L(q) = \int q(\theta) \ln \left(\frac{p(x, y, \theta)}{q(\theta)} \right) d\theta$ for the log model evidence $\ln p(x, y)$. Here, θ can be divided into $J = 3$ groups and $q(\theta) = q_1(v, \alpha)q_2(e)q_3(\rho, \sigma^2)$. We assume that only q_i is unknown among the factorized distributions $\{q_1, q_2, q_3\}$. Then, maximization of the lower bound $L(q)$ is replaced by the one that focuses on q_i , which can be written as

$$\hat{q}_i(\theta_i) = \arg \max_{q_i} \int q_i(\theta_i) \ln \left(\frac{\exp E_i[\ln p(x, y, \theta)]}{q_i(\theta_i)} \right) d\theta_i \quad (10)$$

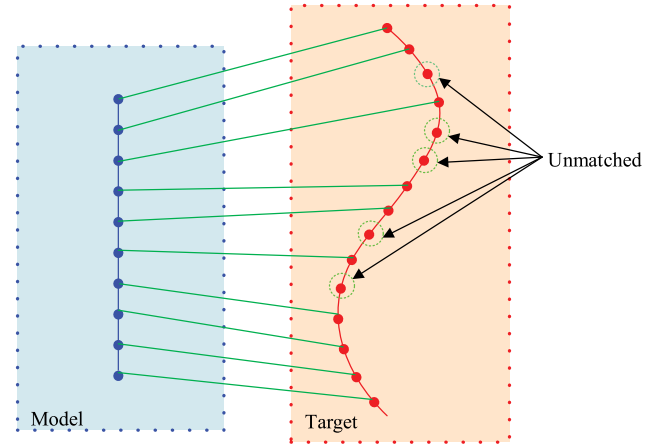


Fig. 6. Correspondence diagram of the registration point.

where $E_i[\ln p(x, y, \theta)] = \int \ln p(x, y, \theta) \prod_{j \neq i} q_j(\theta_j) d\theta_j$. We can obtain the general solution as follows:

$$\ln \hat{q}_i(\theta_i) = E_i[\ln p(x, y, \theta)] + \text{const} \quad (11)$$

The problem, Eq. (9), can be solved by cycle the update q_i with other q_j fixed using Eq. (11) until convergence. The update of q_i is solved by the proposition in the literature [20]. Through the iterative method, the maximum likelihood estimate is continuously approximated to complete the parameter identification. To improve computational efficiency, it is generally necessary to stop iteration after the variance is less than a certain value or the iteration reaches a certain number of times.

After the point set registration is completed, we can implement the mapping between the model point set and the target point set using the transformation parameters (s, R, t) . Hence, we can obtain the target workpiece welding seam trajectory point set from the basic workpiece welding seam trajectory point set by Eq. (3). In addition, the welding seam auxiliary point set of the basic workpiece is not performed for registration, and we generate the target workpiece welding seam auxiliary point set according to the same transformation as the basic workpiece welding seam trajectory point set.

3.3. Curve fitting

The feature point clouds of the basic workpiece and the target workpiece are generated by a certain sampling interval. Besides, the size and deformation degree between the basic workpiece and the target workpiece are not the same. Hence, the model point set and the target point set are not corresponding to each other in the model registration process. Fig. 6 shows the corresponding schematic diagram of the weld point set of the basic model and the target workpiece after registration. As a result, the point set of welding seam trajectory obtained after registration is not smooth enough. In addition, the point set of the welding seam trajectory of the target workpiece has an excessive number of points. Hence, it is necessary to perform curve fitting to make the welding seam trajectory smoother and to enable the welding trajectory planning of the welding robot.

In industrial applications, spline curve fitting and polynomial curve fitting are commonly used curve fitting methods. Compared with polynomial fitting, spline curve fitting has relatively high fitting accuracy and can fit irregular curves, which has better adaptability [21,22]. Hence, we used the B-spline curve fitting method to fit the set of welding seam trajectory points. We describe the welding seam trajectory as an example, since more than one curve needs to be fitted. In general,

the parametric B-spline $Q(t)$ of order k is defined by the following equation

$$Q(t) = \sum_{i=1}^m C_i B_{i,k}(t) \quad (12)$$

where $C_i (i = 1, \dots, m)$ are control points and $B_{i,k}(t)$ are the normalized B-spline functions of order k defined on a knot vector. Here, the k is set to 3 empirically.

We still follow the definition in Section 3.1, and define the welding seam trajectory point set as P . Welding seam trajectory fitting is a continuous adjustment of the control points to make the B spline curve close to the welding seam trajectory point set. The curve fitting process can be defined as an error optimization problem as well. We can define an error function $E(C)$ for the spline control point to detect the error of the fitted curve base on least squares method. The expression of the error function can be written as

$$E(C) = \frac{1}{2} \sum_{i=1}^n d^2(Q(t_i), p_i) \quad (13)$$

where $d(Q(t), p_i) = \min_t \|Q(t) - p_i\|$ is the Euclidean distance of point p_i to the fitting curve $Q(t)$; n is the number of points in point set P . The control points C can be written as a column vector $\hat{C} = [C_1 \ C_2 \ \dots \ C_m]^T$. Similarly, the point set P can be written as $\hat{P} = [p_1 \ p_2 \ \dots \ p_n]^T$. Assuming that the control points are given. Then, the error function $E(C)$ can be rewritten as

$$E(\hat{C}) = \frac{1}{2} \sum_{i=1}^n \left\| \sum_{j=1}^m B_{j,k}(t_i) C_j - p_i \right\|^2 \quad (14)$$

where the $t_i \in [0, 1]$, is scaled sample time. The fitting problem is converted into an error optimization problem, and the control points are obtained when the error function $E(\hat{C})$ is minimized. This can be solved by the partial derivative method. The first-order partial derivative is term of the control point Q_i and can be written as:

$$\frac{dE}{dC_i} = \sum_{a=1}^m \sum_{j=1}^n B_{i,k}(t_a) B_{j,k}(t_a) C_j - \sum_{a=1}^m B_{i,k}(t_a) p_a \quad (15)$$

When we set the Eq. (15) to 0, we can get the equations

$$0 = \sum_{a=1}^m \sum_{j=1}^n B_{i,k}(t_a) B_{j,k}(t_a) C_j - \sum_{a=1}^m B_{i,k}(t_a) p_a \quad (16)$$

$$= H^T H \hat{C} - H^T \hat{P}$$

where $H = [B_{j,k}(t_a)]$ is a matrix with m rows and n columns. Eq. (16) can be rewritten as

$$\hat{C} = (H^T H)^{-1} H^T \hat{P} = [(H^T H)^{-1} H^T] \hat{P} \quad (17)$$

Finally, the control points $\hat{C}$ can be obtained by solving the Eq. (17).

After the welding seam trajectory fitting is completed, the auxiliary welding seam trajectory is fitted by the same method. The purpose of curve fitting is not only to make the trajectory smoother, but also to filter the noise and reduce the trajectory deviation.

3.4. Welding torch pose planning

After the extraction and fitting of the welding seam trajectory, we need to complete the pose planning of the welding torch according to the structure of the workpiece. This is due to the fact that the robot needs not only to acquire the path but also to plan the pose of the welding torch according to the workpiece structure when performing welding tasks. Especially when the workpiece is complex, the pose planning of the welding torch is particularly important. We first calculate the coordinate system of the current welding seam point and then plan the welding torch posture by coordinate transformation.

A diagram of the current welding seam point of the target workpiece is shown in Fig. 7. The welding seam auxiliary point set of the target

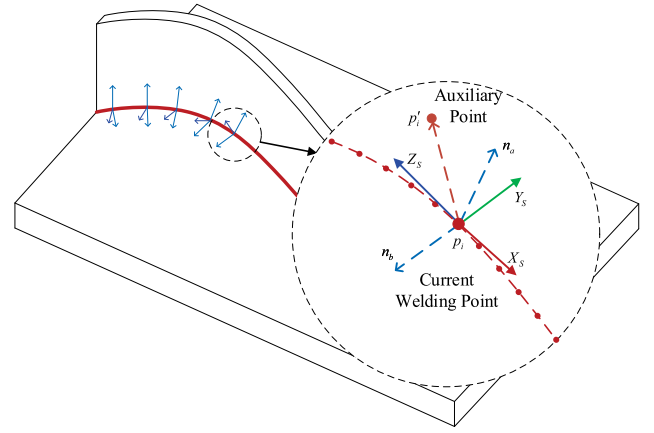


Fig. 7. Schematic diagram of the current welding point of the target workpiece.

workpiece is no longer located on the tangent plane of both sides of the welding seam because it is obtained by a non-rigid transformation of the basic workpiece welding seam auxiliary point set. Therefore, the welding seam auxiliary point set of the target workpiece cannot be directly applied to calculate the welding torch pose directly. And we know the X_s axis direction, the current welding seam point p_i and its auxiliary point p'_i . We need to get the Z_s axis direction. With the same concept as in Section 3.1, we use the approximation method to achieve the acquisition of points in the tangent plane to achieve the solution of Z_s direction. We proposed a new method to obtain the Z_s -axis direction vector by solving for the point composition vector $\vec{n}_a$ and $\vec{n}_b$ in the tangent planes on both sides of the welding seam.

The dense point cloud of the target workpiece is introduced to calculate the Z_s -axis direction vector. The points within the square centered at p_i and with side length $l_w = 10$ mm are the local point set Q . As the direction of X_s axis can be calculated, we can calculate the welding normal plane. Then, We project the local point set Q to the welding normal plane by the following set of equations

$$\begin{cases} \overline{q'_j p_i} \cdot \overline{q_j q'_j} = 0 \\ \overline{p_{i+1} p_i} \cdot \overline{q'_j p_i} = 0 \\ \frac{\overline{q_j q'_j}}{\|\overline{q_j q'_j}\|} = \text{sign}(\overline{q_j q'_j} \cdot \overline{p_{i+1} p_i}) \frac{\overline{p_{i+1} p_i}}{\|\overline{p_{i+1} p_i}\|} \end{cases} \quad (18)$$

where q'_j is the projection point of q_j . To avoid noise, we set a threshold t_p for the projection distance, and retain the projection points whose projection distance is less than the threshold $t_p = 3$ for the local point set Q . The set of projected points is denoted as Q' after filtering. Due to Q' , p'_i and p_i are in the welding normal plane, we divide Q' into two parts Q'_A and Q'_B by a straight line which passes through p'_i and p_i . As an example, the local point set projection at the current welding point of the fillet welding workpiece is shown in Fig. 8.

To find the points in the tangent plane, we take Q'_A as an example. First, we calculate and count the angles consisting of each point in the point set Q'_A with the current welding seam point p_i and the welding auxiliary point p'_i , which is defined as $\angle q'_{a,j} p_i p'_i$. As shown in Fig. 9, we count the histograms of the magnitude of $\angle q'_{a,j} p_i p'_i$. The angle formed by the projection points of the tangent plane is located in the first wave of the histogram. Therefore, we can approximate the unit vector $\vec{n}_a$ from the projection point located in the first wave peak of histogram. Similarly, we can derive the unit vector $\vec{n}_b$ by the same method. Due to the lack of tangent plane point cloud at the beginning and end of the welding seam trajectory due to the structure of the workpiece and the influence of point cloud noise, we cannot directly calculate the Z_s axis

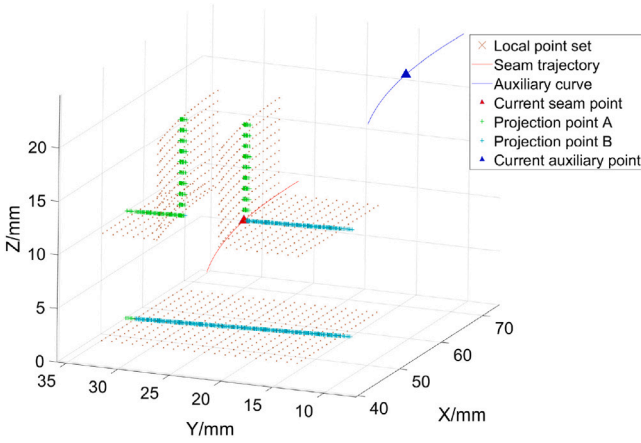
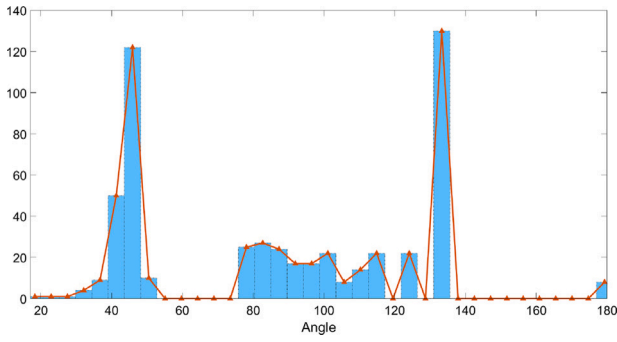


Fig. 8. Local point cloud of the current welding seam point.

Fig. 9. Histograms of the magnitude of $\angle q'_{a,j} p_i p'_i$.

direction. Hence, we first calculate the point located on the Z_S axis by the following equation

$$z_i = R_z * (\bar{n}_a + \bar{n}_b) + p_i \quad (19)$$

where R_z is a constant to increase the distance from z_i to p_i . Empirically, we set R_z to 10.

z_i can be treated as a true auxiliary point to the current welding seam point p_i . The true auxiliary point set Z corresponding to the set of weld trajectory points can be obtained by Eq. (19). As the welding seam trajectory is continuous and smooth, so is the real auxiliary trajectory corresponding to it. Hence, we use the curve fitting method in Section 3.3 to fit the true auxiliary point Z set to filter the noise. Then, the frame of the current welding seam point p_i can be calculated by the following equation

$$\begin{cases} X_S = \frac{\bar{p}_i \bar{p}_{i+1}}{\|\bar{p}_i \bar{p}_{i+1}\|} \\ Z_S = \frac{\bar{p}_i \bar{z}_i}{\|\bar{p}_i \bar{z}_i\|} \\ Y_S = Z_S \times X_S \end{cases} \quad (20)$$

After the current welding point coordinate system is calculated, we can refer to Fig. 4 to complete the calculation of the welding torch posture in the workpiece coordinate system. The conversion between the current welding point coordinate system and the welding torch coordinate system can be achieved by the following transformation matrix.

$$T_T^S = \begin{bmatrix} \sin\omega & \cos\omega & 0 & 0 \\ \cos\omega & -\sin\omega & 0 & 0 \\ 0 & 0 & -1 & l_w \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (21)$$

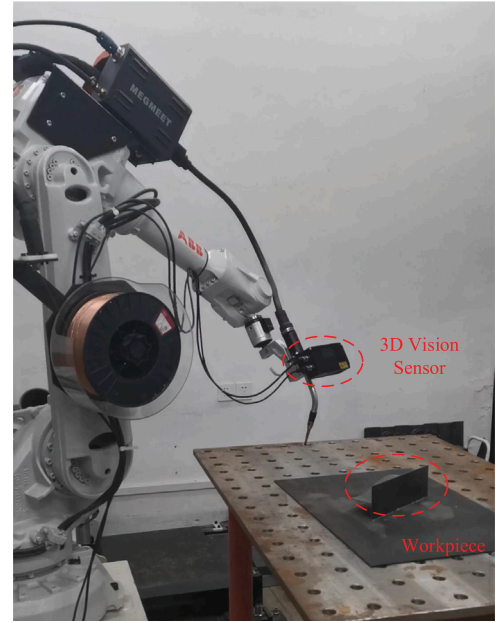


Fig. 10. The welding robot experiment platform.

where l_w is the length of the wire sticking out of the welding torch, generally set to 10 ~ 15 mm. The transformation matrix between the workpiece coordinate system and the welding coordinate system is defined as T_S^W . The transformation matrix between the workpiece coordinate system and the user coordinate system is defined as T_W^U . We can get the expression for the position of the welding torch in the user coordinate system as follows:

$$T_T^U = T_W^U * T_S^W * T_T^S \quad (22)$$

Considering the complexity of the transformation matrix, T_T^U is not formulated in detail here.

4. Experiments and discussion

To verify the effectiveness of the proposed method, offline programming and online programming experiments were conducted. The 3D models of the target workpieces were known in the offline programming experiments. Therefore, the offline programming experiments were verified in the simulation software for welding trajectory planning. In the online programming experiments, the 3D structured light camera was used for the acquisition of the target workpiece point cloud. Both offline and online programming experiments used ABB industrial robots (IRB2600). Fig. 10 shows the welding robot experiment environment of the online programming experiment. In the experiments, the proposed method were run on a win10 system with i7-10875H CPU and 16 GB RAM. The basic model was built in our model generation software based on Visual Studio 2019 and Opencascade.7.6.0. The sampling interval of the 3D model to point cloud conversion was set to 1 mm for both online and offline programming experiments. In the online programming experiments, the point cloud acquired by the 3D vision sensor was sparsely processed using the downsampling method of the voxel grid, and the sampling interval of the voxel grid was also set to 1 mm.

4.1. Offline programming experiment

We selected four types of typical workpieces for offline programming experiments as shown in Fig. 11. These are common fillet welding workpieces, open impeller workpieces, butt welding workpieces and

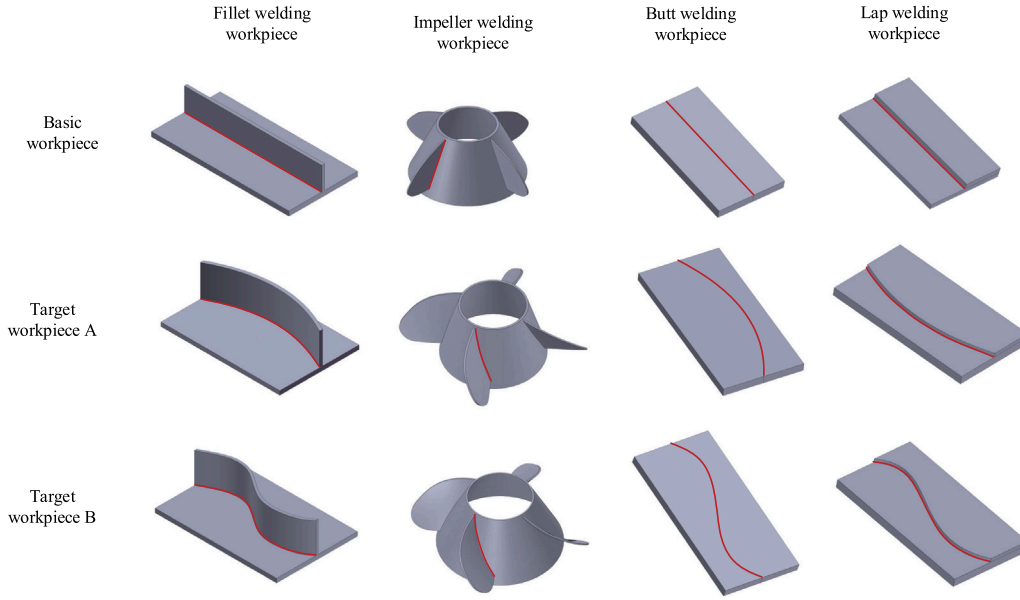


Fig. 11. Typical welding workpieces for off-line programming experiment.

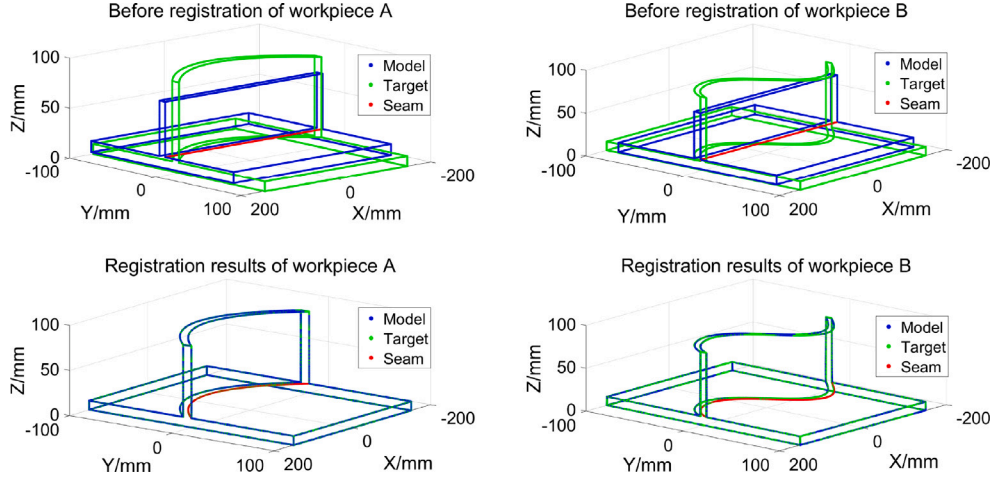


Fig. 12. The registration results of fillet welding workpiece for offline programming experiment.

lap welding workpieces. The four types of workpieces with different deformations were selected as experimental objects to show that the basic model can be applied to workpieces with different deformations. Since the open impeller blades are uniformly distributed. Therefore, we selected only one blade of the open impeller workpiece for the welding trajectory planning in the offline programming experiment. In the offline environment, the 3D models of the basic and target workpieces were generally generated according to the same rules. Therefore, the principles of workpiece coordinate system selection are the same for the basic and target workpieces. It means that the positional deviation of the basic model and the target workpiece point cloud before registration is not significant. As shown in Figs. 12–15, the model point clouds and target point clouds of fillet welding workpiece, open impeller workpiece, butt welding workpiece and lap welding workpiece before registration, respectively.

In the offline programming experiments, the registration parameter β was set to 1.2 for the fillet welding workpieces, 0.9 for the open impeller workpieces and 1.8 for the butt welding workpieces and lap welding workpieces. The rest of the registration parameters were consistent for both workpieces, and the λ was set to 200, the maximum number of iterations was set to 500, and the abort iteration error

threshold was set to $1e-6$. Figs. 12–15 shows the registration results of the fillet welding workpiece open impeller workpiece, butt welding workpiece and lap welding workpiece, respectively.

As shown in Figs. 16 and 17, the welding torch pose planning results of offline programming experiment. Due to the redundancy of the welding torch with two rotational degrees of freedom, the welding torch pose is often adjusted according to the placement of the workpiece in the actual welding task. In the offline programming experiment, the ω were set to 180° , 90° , -60° , 45° for the fillet welding workpieces, the open impeller workpieces, the butt welding workpieces and lap welding workpieces, respectively. The off-line programming experiments of the target welding workpieces are shown in Fig. 18.

4.2. Online programming experiment

To verify that the method proposed in this paper can also be used in online robot programming, we designed online programming experiments. The point cloud of the target workpieces were acquired by a 3D vision sensor (MECH MIND: Mech-Eye NANO), as shown in Fig. 10. The calibration between the vision sensor coordinate system and the robot coordinate system has been completed ahead of schedule. In

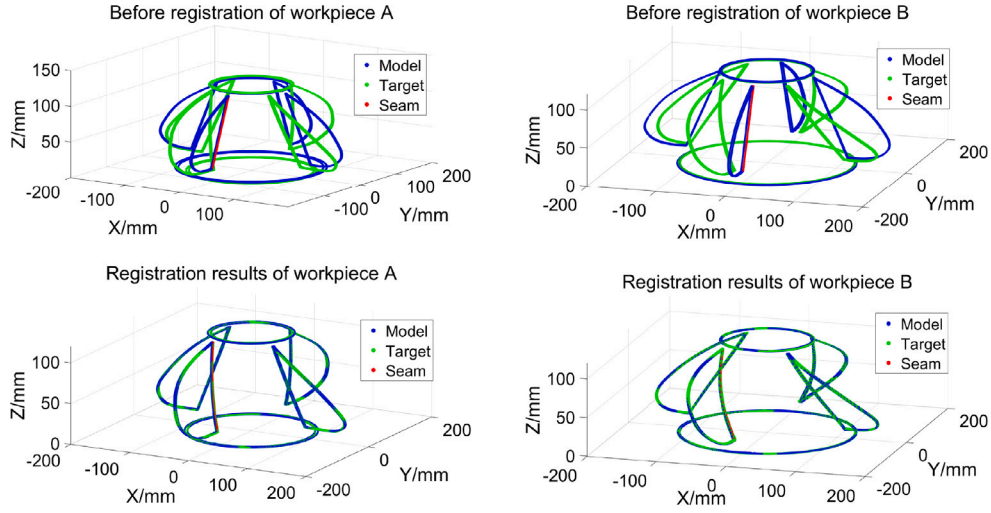


Fig. 13. The registration results of open impeller workpiece for offline programming experiment.

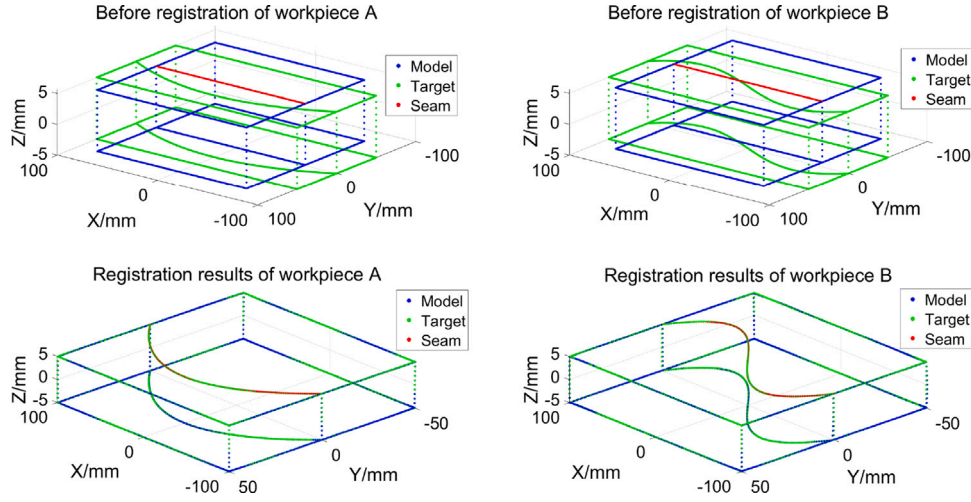


Fig. 14. The registration results of butt welding workpiece for offline programming experiment.

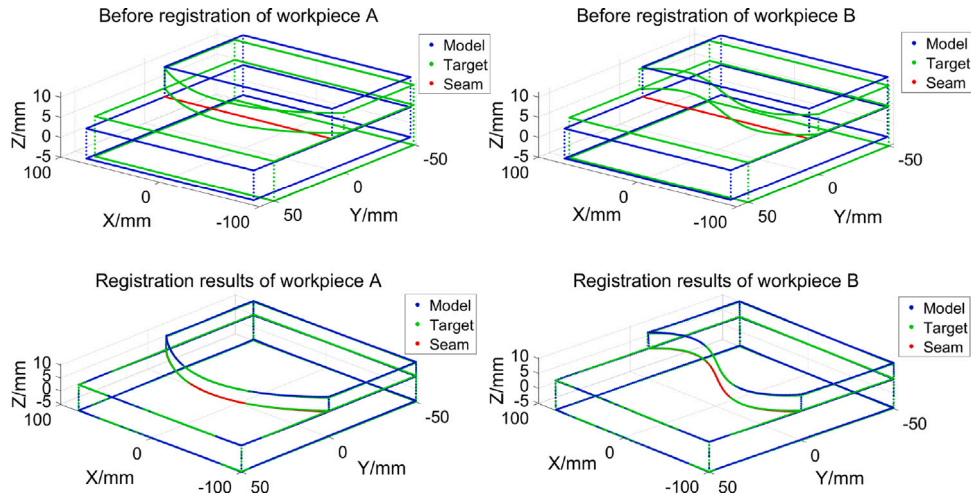


Fig. 15. The registration results of lap welding workpiece for offline programming experiment.

contrast to offline programming experiments where the point cloud of the target workpieces were obtained from the 3D model conversion of the workpiece, the online programming experiments can generally only

obtain a local point cloud of the target workpiece by the vision sensor. A complete point cloud can be obtained for complex workpieces, such as impellers, by taking multiple shots from multiple viewpoints using

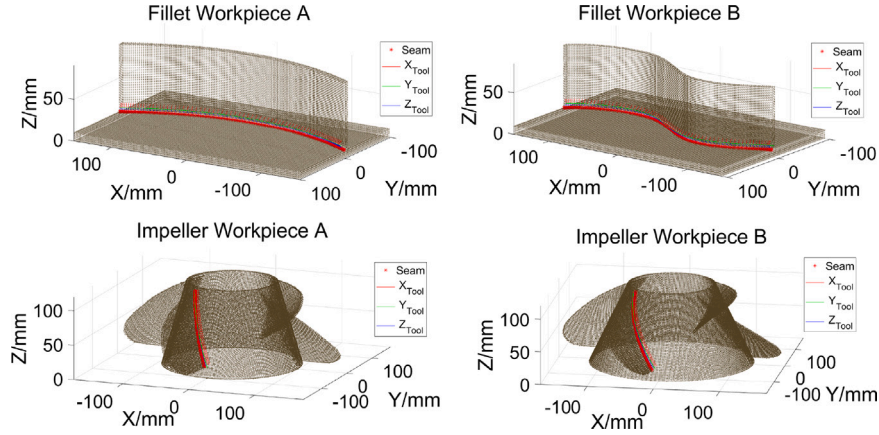


Fig. 16. The welding torch pose planning results of fillet welding and open impeller workpieces in offline programming experiment.

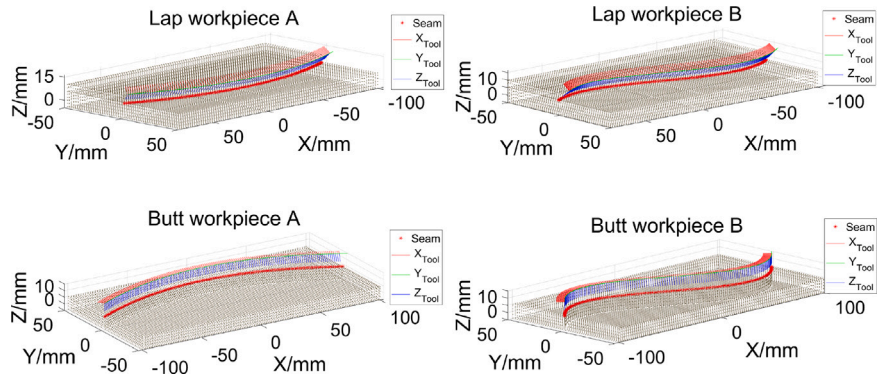


Fig. 17. The welding torch pose planning results of butt welding and lap welding workpieces in offline programming experiment.

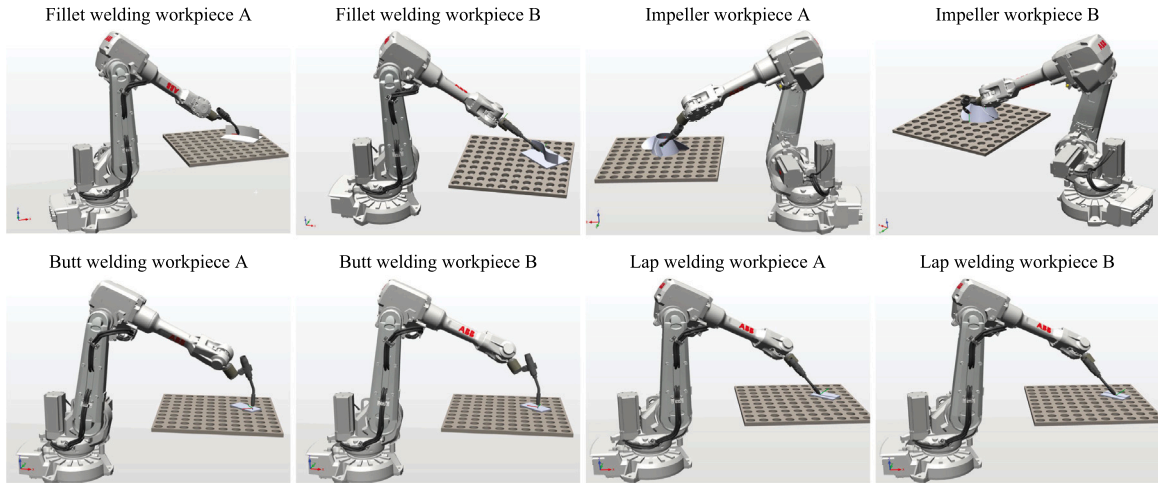


Fig. 18. The off-line programming experiment results of target welding workpieces.

vision sensors. However, multiple shots are not only time-consuming, but also complicated to process and calculate the point cloud obtained from the vision sensor. A single shot of the vision sensor can generally only obtain a localized point cloud of the welding seam, which is the joint of two pieces of metal. The welding seam can be regarded as the intersection line of two surfaces, and the localized point cloud of the welding seam is characterized by a two-sided structure. Therefore, we selected typical fillet welding workpieces, butt welding workpiece and lap welding workpiece as the online programming experimental workpieces, which require only a single shot to obtain the complete welding seam trajectory and its local point cloud without interference.

The fillet welding workpiece, the butt welding workpiece and the lap welding workpiece used in the on-line programming experiments are shown in Fig. 19, where the curved surface deformations in the fillet welding workpieces were made by artificially bending a flat plate manually. Since the proposed method does not involve the study of vision sensor shooting pose planning, we manually adjusted the sensor shooting position to ensure that the point cloud of the target workpiece was clearly captured.

In the online programming experiment, the feature point cloud and dense point cloud of target workpiece were extracted first. To obtain the dense point cloud of target workpiece, the captured point cloud

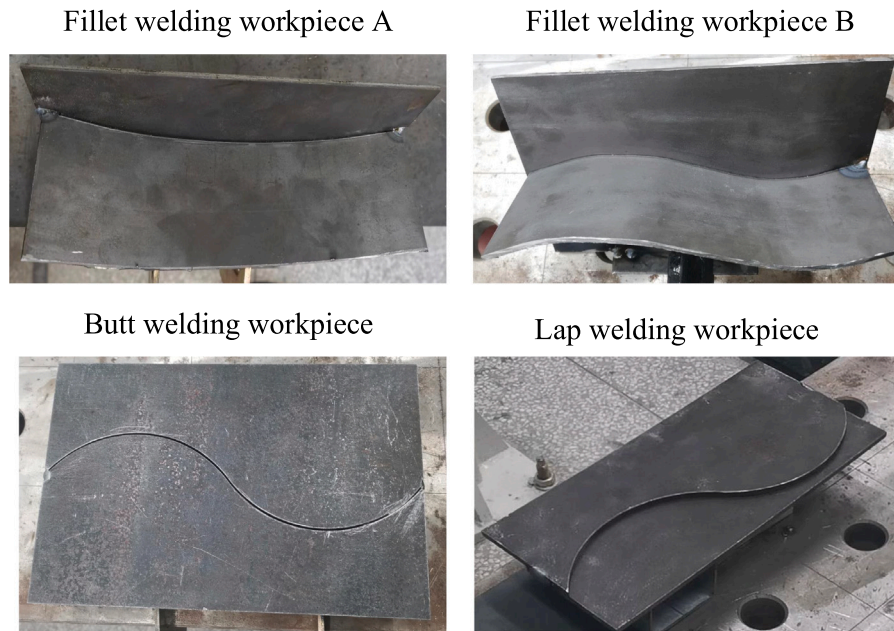


Fig. 19. Typical welding workpieces for online programming experiment.

was first converted to the robot base coordinate system. As shown in Fig. 10, since the Z-axis of the robot's base coordinate system is perpendicular to the working table plane and the height of the working table plane is known, the workpiece and the background point cloud can be segmented by setting a height threshold. Next, the dense point cloud of the target workpiece can be obtained after the point cloud filtering process. To obtain the featured point cloud of the target workpiece, the point cloud edge detection algorithm was applied [23]. Moreover, the target workpieces consists of flat or flat and curved metal plates. We can more easily realized the dense point cloud segmentation, and then obtain the point cloud boundary to realize the feature point cloud extraction.

The registration parameters were set to be consistent with the corresponding welding workpiece in the offline programming experiments. As shown in Figs. 20 and 21, the basic model and target point sets of the fillet welding workpieces, butt welding workpiece and lap welding workpiece for online programming experiment. Compared to the corresponding welding workpiece in the offline programming experiment, the basic model of the welding workpiece in the online programming experiment was generated based on the shooting characteristics of the vision sensor. Besides, the initial positional deviation between the basic model and the target workpiece is larger. As shown in Fig. 22, the welding torch pose planning results of online programming experiments. Similarly, we adjusted the parameter ω in the online programming experiment. In the online programming experiment, the ω were set to 180° for the fillet welding workpieces, butt welding workpieces and lap welding workpieces. To verify the effectiveness of the proposed method, actual welding experiments were performed on fillet welding workpiece B, butt welding workpiece and lap welding workpiece, respectively. The actual welding results are shown in Fig. 23.

4.3. Analysis and discussion

The effectiveness of our proposed method was demonstrated by offline and online programming experiments on typical workpieces. To further evaluate the advantages and disadvantages of the method, We analyzed the welding trajectory planning errors for each workpiece. At first, we analyzed the welding seam trajectory error, the welding torch orientation error and the algorithm operation efficiency generated from the offline and online programming experiments. In addition, we

selected the fillet welding workpiece B as the experimental object and analyzed the effect of registration parameters and point cloud sampling interval on the algorithm. Finally, we performed a comparison of the proposed method with other similar methods.

In the off-line programming experiments, we extract the real welding seam trajectory directly from the 3D model of target workpiece and obtain the real orientation of the welding torch according to the basic model establishing method. In the online programming experiment, we manually obtained the real welding seam trajectory from the target workpiece point cloud and obtained the real orientation of the welding torch by segmenting the point cloud. In the analysis of the welding torch pose, we only analyzed the orientation error γ of the Z_T axis of the welding torch in workpiece frame because the welding torch is a solid of revolution from the perspective of the nozzle. We calculated the maximum error (ME) and root mean square error (RMSE) for each translation error and orientation error, respectively.

The welding seam trajectory error and the welding torch orientation error for each target workpiece of offline and online programming experiments are shown in Table 1. The offline experiment has a relatively small welding seam trajectory error, with the ME not exceeding 0.6 mm and the RMSE less than 0.3 mm. The online programming experiment has a larger welding seam trajectory error than the offline experiment, with the ME not exceeding 1.5 mm and the RMSE less than 0.6 mm. The ME of the welding torch orientation is less than 0.2 rad in both offline and online programming experiments where the ME of the welding torch orientation of the open impeller workpiece reaches 0.193 rad. The main reason is the complex shape of the impeller workpiece, especially the insufficient local point cloud of the impeller blade near the start and end of the welding seam trajectory, affected the calculation of the welding torch orientation. The maximum RMS value of the welding torch orientation error is 0.057 rad for the fillet welding workpiece B in the online programming experiment. To some extent, we can regard the fillet welding workpiece B in the online programming experiment as the result of adding point cloud noise to the offline programming experiment, even though the degree of deformation of the target fillet welding workpiece is different between the online and offline programming experiments. The experimental results in Table 1 show that the proposed method can not only effectively obtain the welding seam trajectory of the target workpiece and performing welding torch pose planning.

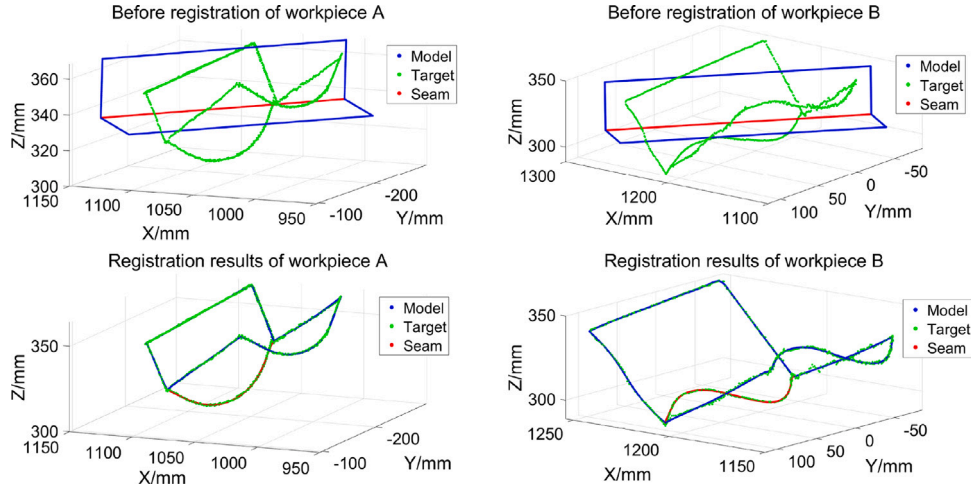


Fig. 20. The registration results of fillet welding workpieces for online programming experiment.

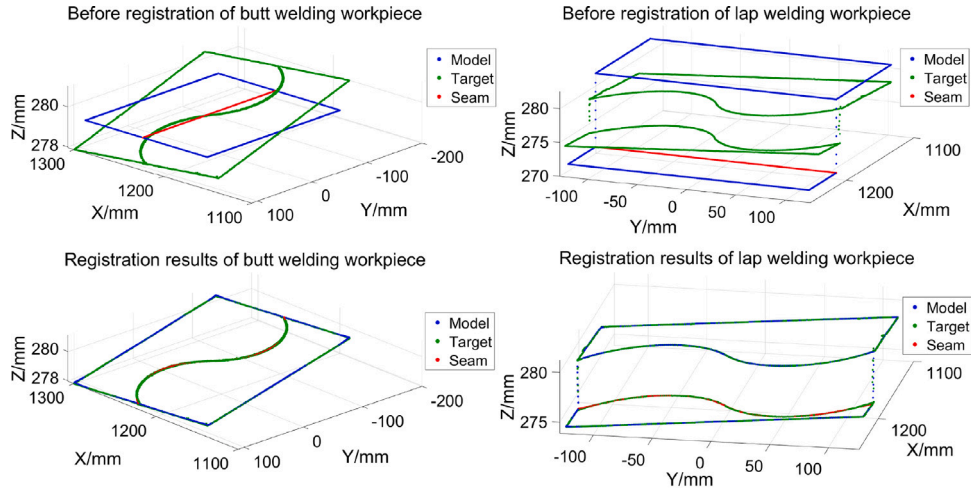


Fig. 21. The registration results of butt welding and lap welding workpiece for online programming experiment.

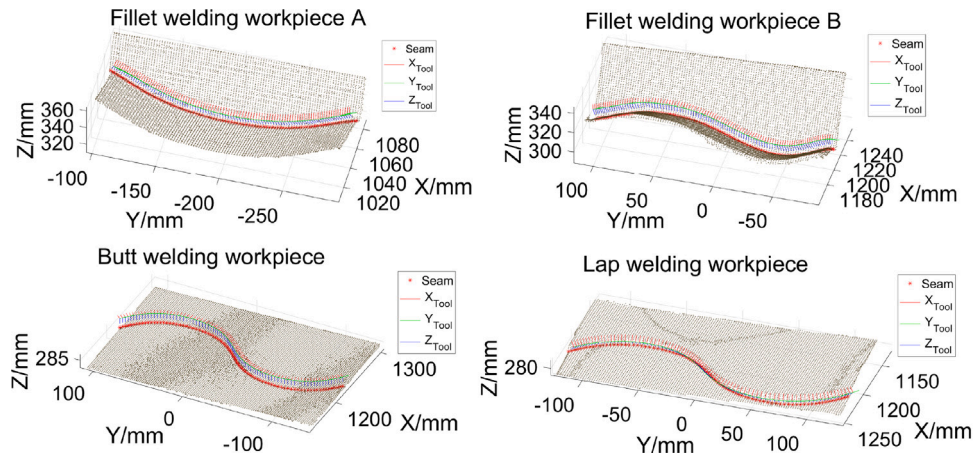


Fig. 22. The welding torch pose planning results of online programming experiment.

We measured the computing time for both offline and online programming experiments. The number of point sets and the computing time for offline and online programming experiments are shown in Table 2, where the preprocessing is the processing time for converting

the 3D model to a point cloud or for the vision sensor to acquire a point cloud of the workpiece. The registration computing consumed the largest proportion of time in the experiment, especially the registration process of the open impeller workpiece B took the longest time of



Fig. 23. The actual welding results for online programming experiment.

Table 1

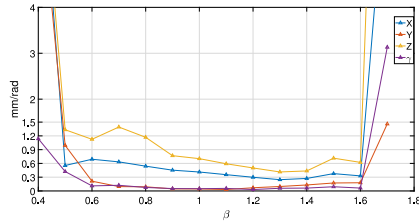
The welding seam trajectory error and welding torch orientation error of experiments.

Scenario	Workpiece	Translate error (ME/RMSE)			Orientation error (ME/RMSE)
		X (mm)	Y (mm)	Z (mm)	
Off-line	Fillet welding workpiece A	0.238/0.13	0.421/0.186	0.062/0.017	0.03/0.008
	Fillet welding workpiece B	0.298/0.11	0.501/0.255	0.069/0.018	0.03/0.01
	Impeller workpiece A	0.436/0.153	0.374/0.14	0.374/0.121	0.135/0.026
	Impeller workpiece B	0.559/0.176	0.356/0.13	0.244/0.101	0.193/0.035
	Butt welding workpiece A	0.49/0.17	0.45/0.16	0.23/0.103	0.035/0.012
	Butt welding workpiece B	0.34/0.15	0.515/0.23	0.158/0.095	0.034/0.012
	Lap welding workpiece A	0.63/0.143	0.462/0.169	0.183/0.097	0.023/0.009
	Lap welding workpiece B	0.27/0.136	0.526/0.165	0.234/0.073	0.023/0.009
On-line	Fillet welding workpiece A	1.475/0.412	1.024/0.271	1.366/0.374	0.133/0.063
	Fillet welding workpiece B	1.383/0.372	1.238/0.385	1.472/0.457	0.123/0.057
	Butt welding workpiece	1.293/0.253	1.356/0.343	1.439/0.417	0.115/0.045
	Lap welding workpiece	1.348/0.325	1.462/0.423	1.243/0.221	0.103/0.043

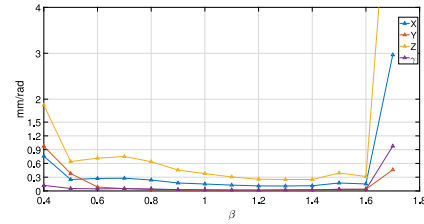
Table 2

The number of point sets and the computing time for offline and online programming experiments.

Scenario	Workpiece	Number of point set		Running time (ms)				
		Model	Target	Pre-processing	Registration	Curve fitting	Pose planning	Total
Off-line	Fillet welding workpiece A	6829	7280	3227	3100	25	658	7010
	Fillet welding workpiece B	6829	7352	4326	3300	31	789	8446
	Impeller workpiece A	9988	9896	10 518	12 450	49	980	23 997
	Impeller workpiece B	9988	10 760	10 735	15 290	58	1110	27 193
	Butt welding workpiece A	3355	4288	3145	3630	45	695	7515
	Butt welding workpiece B	3355	4360	3236	4740	42	673	8691
	Lap welding workpiece A	4315	4272	3328	4190	54	693	8265
	Lap welding workpiece B	4315	4280	3145	3990	63	739	7937
On-line	Fillet welding workpiece A	2611	1152	3856	3150	22	213	7241
	Fillet welding workpiece B	2611	1006	4172	2990	18	205	7385
	Butt welding workpiece	1119	1560	2582	3020	25	195	5822
	Lap welding workpiece	927	1418	2670	1400	27	228	4325



(a) Max error



(b) Root mean square error

Fig. 24. Effect of registration parameter β on welding seam trajectory and torch orientation.

15290 ms. One of the reasons for the long registration time of the open impeller is the relatively large number of registration point sets. In addition, the other main reason is that all four blades of the open impeller workpiece are deformed to some extent, resulting in a relatively slow convergence of the registration algorithm. The comparison between the fillet welding workpiece and the open impeller workpiece in the offline experiment showed that the more complex the structure, the greater the deformation, leading to an increase in the registration time to some extent.

We analyzed the effect of registration parameters and point cloud sampling interval on the accuracy of the proposed algorithm using fillet welding workpiece B in offline programming experiment as the experimental object for further analysis. Among the registration parameters we mainly analyzed the effect of β and λ on the welding seam trajectory and the welding torch orientation. The experimental settings, except for the parameters to be analyzed, were the same as in the off-line programming experiments with fillet welding workpiece B. As shown in Figs. 24 and 25, the ME and RMSE of the welding seam trajectory

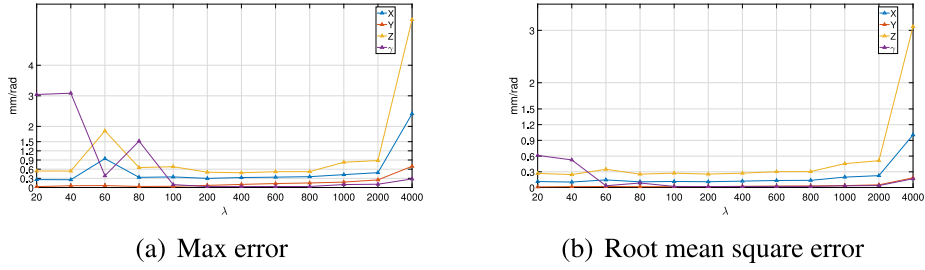
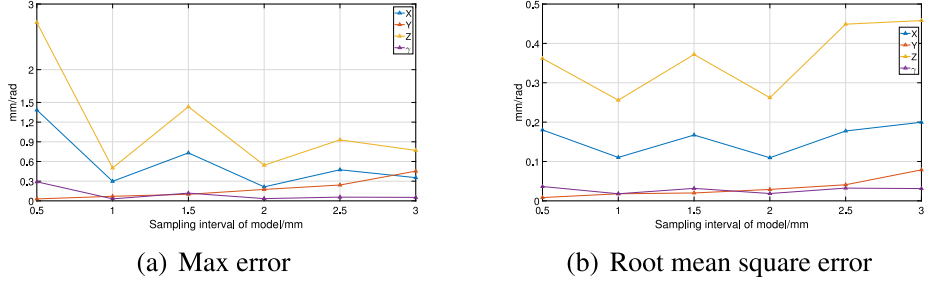
Fig. 25. Effect of registration parameter λ on welding seam trajectory and torch orientation.

Fig. 26. Effect of model sampling interval on welding seam trajectory and torch orientation.

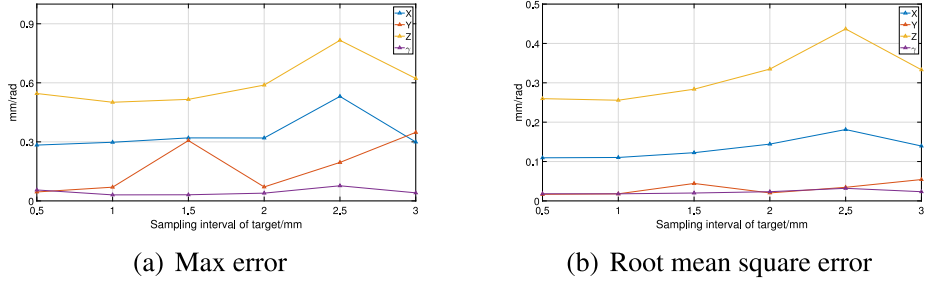


Fig. 27. Effect of target sampling interval on welding seam trajectory and torch orientation.

and welding torch pose for different values of β and λ , respectively. The values of β and λ both obtained high registration accuracy within a certain range. Compared with β , λ took a larger range of values and the error fluctuation was not significant. Overall, the welding torch pose error increases as the welding seam trajectory deviation increases. To further analyze the effect of the point cloud sampling interval on the welding seam trajectory and welding torch pose, we conducted experiments with different sampling interval settings for the basic model and the target workpiece, respectively. As shown in Figs. 26 and 27, the welding seam trajectory error and welding torch pose error are shown for the basic model and the target workpiece at different sampling intervals, respectively. The RMSE of the obtained welding seam trajectory varies little with different point cloud sampling intervals for the base model and the target workpiece. In addition, the accuracy of the welding seam trajectory and welding torch orientation does not increase as the sampling interval decreases. In the experiment, the length of the welding seam trajectory in the basic workpiece is 300 mm, and the length of the welding seam trajectory in the target workpiece is 327 mm. We found that when the sampling interval is such that the ratio of the number of points in the model welding seam trajectory point set and the target welding seam trajectory point set is close to 0.5 or 1, the obtained welding seam trajectory error is smaller.

The main advantage of the method proposed in this paper, which uses the point cloud as a kernel, is that it provides a general method for solving repetitive programming of the same type of welding workpiece. Considering the inconsistency of the experimental running platform as well as the experimental workpieces, we mainly performed a functional

comparison of similar studies. The comparison results are shown in Table 3. Compared to other methods, the proposed method enables not only path extraction but also welding torch pose planning, and can be applied in both offline and online programming environments. Besides, we adopted the comparison method in the literature [8] by selecting fillet welding workpiece B and impeller workpiece B from the offline experiments and the fillet welding workpiece B from the online experiments as the objects to highlight the programming efficiency of the proposed method. The offline programming experiments were performed in the ABB Robotics simulation software. The comparison results shown in Table 4, the proposed method can improve the programming efficiency of welding robots. The final offline and online programming experimental results demonstrate the effectiveness of the proposed method.

5. Conclusion

In this paper, we propose a novel model-based welding trajectory planning method for identical structural workpieces to simplify the welding trajectory planning of identical structural workpieces and improve the programming efficiency of welding robots. The final offline programming experiments and online programming experiments demonstrate the effectiveness of the proposed method. The conclusions are summarized as follows:

A new model-based welding trajectory planning method can not only realize the extraction of the welding seam trajectory of the identical structural workpieces, but also realize the pose planning of the welding torch through the dense point cloud of the target workpiece.

Table 3

The function comparison results of similar methods.

Method	Path extraction	Torch pose planning	Workpiece type	Online or offline programming
Ref. [24]	✓	×	Cylinder and Y-shaped workpieces	Online
Ref. [25]	✓	✓	Flat	Online
Ref. [26]	✓	×	Flat	Offline
Ours	✓	✓	Flat and open impeller	Online and offline

Table 4

The efficiency comparison results of programming methods.

Method	Fillet welding workpiece B in offline programming	Impeller workpiece B in offline programming	Fillet welding workpiece B in online programming
Teach-in programming	/	/	About 1200 s
Off-line programming software	About 720 s	About 900 s	/
Proposed method	8.446 s	27.193 s	7.385 s

The proposed method can be applied to the planning of welding seam trajectories for complex structural workpieces by creating a corresponding simplified basic model of the workpiece.

Experimental verification shows that the proposed method can be used for both offline programming and online programming for welding robots.

It is worth noting that we have described in the paper how to build the basic model. However, it is not always necessary to select a basic workpiece to build the basic model. Complex workpieces can also be used as basic models for welding simple or more complex workpieces.

CRediT authorship contribution statement

Weihua Fang: Conceptualization, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xincheng Tian:** Investigation, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.rcim.2024.102772>.

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